

Solution Guide – Telecommunication Churn Analysis Using EXCEL

1. Basic-Level Questions and Answers

Q1. How many customers have churned, and what is their proportion compared to the total customer base?

Answer: According to the earlier workbook results, 598 customers churned out of 4,250 customers, giving a churn rate of 14.07%.

This means approximately 14 out of every 100 customers left the service. Airtel should monitor this rate over time and develop customer retention strategies to reduce customer loss.

Q2. What is the average account length for churned customers compared to non-churned customers? Does the account tenure influence churn?

Answer: Churned customers have an average account length of 95.42, compared with 102.72 for non-churned customers.

This suggests that churned customers have shorter account lengths on average. However, account length alone cannot establish the reason for churn. Airtel should also consider customer service calls, plan types, and usage charges.

Q3. How does the subscription to international plans correlate with customer churn?

Answer: The reported churn rate is 40.90% for customers with an international plan and 8.90% for customers without one.

This difference suggests a strong association between international-plan subscription and churn in the analyzed data. Airtel should review international calling prices and plan benefits to determine whether customers' needs are being met.

Q4. Is there a significant difference in the number of customer service calls between churned and non-churned customers?

Answer: Churned customers make an average of 1.65 customer service calls, while non-churned customers make an average of 1.14 calls.

This suggests that churned customers contact customer service more frequently. Repeated calls may indicate unresolved problems or dissatisfaction, so Airtel should investigate common complaints and improve resolution times.

Q5. Do day, evening, and night call charges significantly differ between churned and non-churned customers?

Answer: The earlier analysis indicates that churned customers generally have higher day, evening, and night charges.

Higher charges may be associated with customer dissatisfaction, especially when customers believe their plans are expensive. Airtel should compare average charges across both groups and review pricing options for high-usage customers. The available figures do not establish statistical significance on their own.

Q6. What is the relationship between total day minutes and churn?

Answer: Churned customers use an average of 201.18 daytime minutes, compared with 174.11 minutes for non-churned customers.

This suggests that churned customers have higher daytime usage on average. Airtel could evaluate whether affordable plans for frequent daytime callers would improve customer satisfaction and retention.

Q7. How does having a voice mail plan affect churn rates?

Answer: Customers with a voicemail plan have a reported churn rate of 7.37%, compared with 16.44% for customers without one.

Customers with voicemail plans are less likely to churn in these results. Airtel could investigate whether additional features increase customer satisfaction and consider how similar benefits could be offered to other customer segments.

Q8. Analyze the total international minutes and charges for churned vs. non-churned customers. Are higher usage and charges associated with higher churn?

Answer: The earlier analysis indicates that churned customers generally have higher international calling minutes and charges.

This suggests that international usage and associated costs may be relevant to customer retention. Airtel should compare the exact average international minutes and charges for both groups before concluding how strong the relationship is.

Q9. Is there a correlation between the number of total day calls and churn?

Answer: The earlier results indicate that churned customers make slightly fewer daytime calls on average, despite having higher average daytime minutes.

This suggests that call frequency and total call duration may show different patterns. Airtel should calculate the correlation between daytime call counts and the churn indicator to determine the direction and strength of the relationship.

Q10. Compare the average total evening charges for churned vs. non-churned customers. Is there a significant difference?

Answer: The reported average evening charge is 17.47 for churned customers and 17.08 for non-churned customers.

The difference is relatively small, although churned customers have a slightly higher average. A statistical test would be required to determine whether this difference is significant or could have occurred by chance.

2. Medium-Level Questions and Answers

Q1. Perform a segmented analysis of churn by area code. Is there a particular area with a significantly higher churn rate?

Answer: The earlier workbook results report that area code 510 has the highest churn rate at approximately 15.06%.

This indicates that churn may vary across geographic areas. Airtel should compare the churn rate and customer count for every area code, then investigate possible differences in service quality, pricing, and customer needs. A statistical test would be needed to establish whether the difference is significant.

Q2. Explore the relationship between the number of customer service calls and churn, considering different thresholds (e.g., more than 3 calls).

Answer: Customers who make more than three customer service calls have a reported churn rate of approximately 50.75%.

This suggests that repeated customer service calls are associated with a higher risk of churn. Airtel should identify customers who contact support frequently and provide timely assistance to resolve their issues before they decide to leave.

Q3. Investigate the correlation between account length and churn. Are newer customers more likely to churn?

Answer: Churned customers have a lower average account length than non-churned customers, based on the earlier results.

This suggests that customers with shorter account histories may be more likely to leave. Airtel should divide customers into account-length groups and compare their churn rates to identify whether newer customers need better onboarding, support, or loyalty benefits.

Q4. Analyze the impact of international call charges on churn among subscribers of international plans.

Answer: The earlier analysis suggests that customers with international plans have a higher churn rate than customers without them. However, this alone does not establish whether international charges are responsible for churn among international-plan subscribers.

Airtel should analyze only customers with international plans, divide them into high- and low-charge groups, and compare their churn rates. This would help determine whether international pricing may be a factor in customer departures.

Q5. Examine how evening usage patterns (calls and charges) relate to customer churn.

Answer: The reported churn rate for the high-evening-usage group is 16.21%, compared with 11.89% for the low-evening-usage group.

This suggests that customers with higher evening usage may be more likely to churn. Airtel should analyze evening minutes, call counts, and charges together to understand whether usage patterns or pricing differences are associated with customer loss.

Q6. Perform a cluster analysis to identify distinct customer groups based on service usage patterns (day, evening, night, international).

Answer: The earlier analysis identified a heavy-user cluster with a reported churn rate of approximately 14.12%.

Clustering can help group customers with similar daytime, evening, nighttime, and international calling behavior. Airtel can compare churn rates across the resulting clusters and develop targeted offers for higher-risk groups. The cluster composition and results should be validated before making business decisions.

Q7. How does the presence of a voice mail plan interact with the number of voice mail messages in relation to churn?

Answer: The earlier results show lower churn among customers with voicemail plans than among customers without them.

To answer this question fully, customers with voicemail plans should be grouped by the number of voicemail messages they receive, and the churn rate of each group should be calculated. This would help determine whether voicemail usage is associated with customer retention. The earlier summary does not provide the exact churn rates for these message-volume groups.

Q8. Investigate the relationship between total day minutes and total day charges with churn. Is there a point of inflection where higher usage leads to higher churn?

Answer: Churned customers have higher average daytime usage than non-churned customers: 201.18 minutes compared with 174.11 minutes.

This suggests a relationship between daytime usage and churn, but the averages alone cannot identify a specific point at which churn increases sharply. Airtel should plot daytime minutes against charges, divide customers into usage groups, and compare their churn rates to identify any potential threshold.

Q9. Does the number of total international calls affect churn differently for customers with and without international plans?

Answer: The earlier analysis indicates that customers with international plans have a higher overall churn rate.

However, this does not establish how the number of international calls affects churn separately for customers with and without those plans. Airtel should divide customers by international-plan status, group them by international call counts, and compare churn rates within each group.

Q10. Explore the effect of different customer service call reasons on churn. Are certain types of calls more indicative of churn risk?

Answer: The available data dictionary does not include a customer service call-reason field. It only records the number of customer service calls made by each customer.

Therefore, the dataset cannot identify which specific complaint types are associated with higher churn. Airtel should collect additional information about complaint categories, such as billing, network issues, and plan changes, to analyze their relationship with customer departures.

3. Advanced-Level Questions and Answers

Q1. Develop a predictive model to identify the likelihood of churn based on customer usage patterns and demographic information. Exclude machine learning models for simplicity.

Answer: Logistic regression can be used to estimate the probability that a customer will churn based on variables such as account length, international-plan status, customer service calls, and usage charges.

The earlier analysis reported a logistic regression accuracy of approximately 81.97%. The model can help identify customers who may be at risk of leaving so Airtel can prioritize retention efforts. Accuracy alone is not enough to evaluate a churn model; precision, recall, confusion matrix results, and validation performance should also be reviewed.

Q2. Conduct a time series analysis to identify trends and seasonality in churn rates.

Answer: Time series analysis requires churn observations linked to dates or time periods.

The available data dictionary does not list a date or cancellation-time field. Therefore, reliable monthly trends and seasonal patterns cannot be established from the described dataset alone. Airtel should collect dated customer and churn records to analyze changes over time and plan retention campaigns accordingly.

Q3. Use cluster analysis to segment customers into distinct groups based on their service usage and demographic profiles. Identify which segments are more prone to churn.

Answer: Cluster analysis can group customers according to their service usage and demographic characteristics, helping identify distinct customer segments.

The earlier analysis reported a heavy-user cluster with a churn rate of approximately 14.12%, although this result alone does not show that heavy users are the highest-risk segment. Airtel should compare churn rates across all clusters and identify which groups need targeted retention campaigns.

Q4. Analyze the interaction effects between various customer attributes (e.g., international plan and total international minutes) on churn.

Answer: Interaction analysis examines whether the combined effect of two customer attributes is different from their individual effects.

For example, Airtel could examine whether customers who have an international plan and make many international calls have a different churn rate from customers who have only one of these characteristics. This analysis can help identify specific customer groups that may need more suitable international plans. The earlier summary reports a churn rate of 42.35%

for customers with an international plan and high international usage, but a formal interaction analysis would be needed to establish the combined effect statistically.

Q5. Evaluate the economic impact of churn by calculating the potential revenue loss associated with churned customers. Consider factors such as average revenue per user (ARPU) and customer lifetime value (CLV).

Answer: The earlier workbook results report approximately 39,188.24 in revenue associated with churned customers and around 3.56 million in potential customer lifetime value (CLV).

These estimates indicate that customer departures may have a meaningful financial impact. Airtel can use revenue and CLV estimates to prioritize retention efforts for valuable at-risk customers. However, the calculations and assumptions should be checked against the dataset before the figures are treated as confirmed financial losses.